Risk Analysis and Management

Risks

- Risks are potential problem/uncertainty that might affect the successful completion of a software project.
- Risk analysis and management are intended to help a software team understand and manage uncertainty during the development process.
- The work product is called a Risk Mitigation, Monitoring, and Management Plan (RMMM).

Risks

- Two risk Strategies :

1. Reactive strategy

   Software team does nothing till the risk becomes real.

2. Proactive strategy

   Risk management begins long before technical work starts.

   Risks are identified and prioritized by importance. Then team builds a plan to avoid risks if they can or minimize their probability of occurrence or establish plan if risks become real.

Risks

- Categories of risks

1. Project risks
   - Threaten the project plan.
   - If project risk becomes real, it is likely project schedule will slip and the costs will increase
   - Identifies problems related to budgetary, schedule , personnel and resource

2. Technical risk
   - Threaten the quality of the software to be produced
   - Identifies problems related to design, implementation, maintenance etc

Risks

3.Business risk

- Threatens the viability of the software to be built.
- Eg of business risk

1. Building an excellent product that no one wants.
2. Building a product that no longer fits into the overall business strategy.
3. Building a product that the sales force do not how to sell.
4. Change of management
5. Losing budgetary

Risk Identification

- Steps for risk analysis

1. **Risk Identification**

- Lists the risks associated with a specific project.
- Use a risk item checklist to identify risks

  Risk item check list – A set of questions relevant to each risk.

A project manager attains a feeling of staffing risk by answering the following set questions---→

1. Are enough people available ?
2. Are the best people available ?
3. Will the project team members be working full time / part time on the project ?
4. Have the staff members received necessary training?

No. of negative responses to the questions $\propto$ degree to which the project is at a staffing risk

Risk projection

2. **Risk projection**

- Also called risk estimation.
- Rates each risk in 2 ways :

1. Find the probability of occurrence of each risk.
2. Find the impact of the problems associated with each risk.

Step 1 : Prepare a risk table.

a) Lists all risks identified in the first column.

b) Category of the risk is identified in the second column.

PS – project size risk.

TE – Technical risk.

BU – Business risk.

Risk projection

c) Probability of occurrence is entered in the third column.

Individual team members are polled for their estimates in a round robin fashion until a single consensus is obtained.

d) Impact of each risk on the 4 risk components is assessed .

4 risk components are

1. Performance

If the risk becomes real will the requirements be met and will the product be fit for its intended use?

2. Cost

If the risk becomes real will the project budget be maintained?

3. Support

If the risk becomes real will the resultant software be easy to correct , adapt and enhance?

Risk projection

4. Schedule

Will the project schedule be maintained and the product will be delivered on time if the risk becomes real?

The impact of each risk on the 4 components can be categorized as

1- Catastrophic
2- Critical
3- Marginal
4- Negligible

The values are averaged to determine an overall impact value.

Risk projection

Risks	Category	Probability	Impact	RMMM
Size estimate may be significantly low	PS	60%	2	
Larger number of users than planned	PS	30%	3	
Less reuse than planned	PS	70%	2	
End-users resist system	BU	40%	3	
Delivery deadline will be tightened	BU	50%	2	
Funding will be lost	CU	40%	1	
Customer will change requirements	PS	80%	2	
Technology will not meet expectations	TE	30%	1	
Lack of training on tools	DE	80%	3	
Staff inexperienced	ST	30%	2	
Staff turnover will be high	ST	60%	2	
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Impact values:
1—catastrophic
2—critical
3—marginal
4—negligible

Table prior to sorting

Risk projection

- Next the table is sorted by high probability and high impact.
- Project manager defines a cut off line implying only risks above the line will be managed.
- The column labeled RMMM contains a pointer into a Risk Mitigation Monitoring and Management Plan(A collection of risk information sheets developed for all risks that lie above the cutoff line)

Risk projection

Step 2 : Assessing the risk impact

a) Estimate the cost of each risk in the table.

- How to estimate the cost of each risk ?

Eg :

Assume that the software team defines a project risk in the following manner:

Risk identification. Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.

Risk probability. 80% (likely).

Risk impact. 60 reusable software components were planned. If only 70 percent can be used, 18 components would have to be developed from scratch (in addition to other custom software that has been scheduled for development). Since the average component is 100 LOC and local data indicate that the software engineering cost for each LOC is $14.00, the overall cost (impact) to develop the components would be **18 x 100 x 14 = $25,200**.

Risk projection

b) Calculate the risk exposure foe each risk in the table ----→

Risk exposure. *RE* = Prob. Of occurrence x cost of risk

=0.80 x 25,200 ~ $20,200.

c) Calculate the total exposure for all risks(above the cutoff line)

The total risk exposure provides a means for adjusting the final cost estimates for a project.

If RE is greater than 50 % of the project cost , the viability of the project must be evaluated

Risk Refinement

3. Risk Refinement

- During early stages of project planning, a risk may be stated quite generally.
- As time passes and more is learned about the project and the risk, it may be possible to refine the risk.
- One way to do this is to represent the risk in condition-transition-consequence (CTC)
- Given that <condition> then there is concern that (possibly) <consequence>.

Risk Refinement

- Using the CTC format for the reuse risk ----→
- Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 % of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 % of components.
- This general condition can be refined in the following manner:

Sub condition 1

Certain reusable components were developed by a third party with no knowledge of internal design standards.

Sub condition 2

Certain reusable components have been implemented in a language that is not supported on the target environment.

Risk Mitigation Monitoring and Management

4. **Risk Mitigation Monitoring and Management**

All the risk analysis activities presented so far have single goal i.e. to assist the project team in developing a strategy to deal with risk.

An effective strategy is--→

1. **Risk Mitigation (avoidance)**

For example,

assume Risk -High staff turnover is noted as a project risk,

Prob. Of occurrence -70% (high) and\

Impact-2 (critical)

RMMM

To mitigate this risk, project management must develop a strategy for reducing turnover.

Among the possible steps to be taken --→

- Meet with current staff to determine causes for turnover (e.g., poor working conditions, low pay, and competitive job market).
- Mitigate those causes that are under our control before the project starts.
- Organize project teams so that information about each development activity is widely dispersed.
- Define documentation standards and establish mechanisms to be sure that documents are developed in a timely manner.
- Assign a backup staff member for every critical technologist.

RMMM

2. **Risk monitoring**

- As the project proceeds, risk monitoring activities commence.
- The project manager monitors factors that may provide an indication of whether the risk is becoming more or less likely.

In the case of high staff turnover, the following factors can be monitored:

- General attitude of team members based on project pressures.
- The degree to which the team has jelled.
- Interpersonal relationships among team members.
- Potential problems with compensation and benefits.
- The availability of jobs within the company and outside it.

RMMM

- Also the project manager should monitor the effectiveness of risk mitigation steps.

- The project manager should monitor documents carefully to ensure that each can stand on its own and that each imparts information that would be necessary if a newcomer were forced to join the software team somewhere in the middle of the project.

RMMM

3. Risk management

- Assumes that mitigation efforts have failed and that the risk has become a reality (a number of people announce that they are leaving)
- If the mitigation strategy has been followed, backup is available, information is documented, and knowledge has been dispersed across the team.
- Those individuals who are leaving are asked to stop all work and spend their last weeks in “knowledge transfer mode. This might include video-based knowledge capture, the development of “commentary documents,” and/or meeting with other team members who will remain on the project.

RMMM Plan/Risk Information sheet (RIS)

- The RMMM plan documents all work performed as part of risk analysis and are used by the project manager as part of the overall project plan.

- Some software teams do not develop a formal RMMM document. Rather, each risk is documented individually using a risk information sheet (RIS)

- Once RMMM has been documented and the project has begun, risk mitigation and monitoring steps commence. And if the risk becomes real then the project manager applies the management steps.

Example of an RIS

Risk ID: P02-4-32 **Date :** 5/9/02 **Prob:** 80% **Impact :** High

Description :

Only 70 % of the software components scheduled for reuse will be integrated into the application. The remaining functionality will have to be custom developed.

Refinement/Context

Sub condition 1

Certain reusable components were developed by a third party with no knowledge of internal design standards.

Sub condition 2

Certain reusable components have been implemented in a language that is not supported on the target environment.

Mitigation/Monitoring :

1. Contact third party to determine conformance with design standards.
2. Check to determine if language support can be acquired.

Example of an RIS

Management Plan :

- RE computed to be $20,200. Allocate this amount within project cost.
- Develop revised schedule assuming 18 additional components will have to be custom built.
- Allocate staff accordingly,

Current Status :

5/12/2002 : Mitigation steps initiated.

Originator : Mr. ABC

Assigned: Mr. CDE

RMMM Document

Risk information sheet			
Risk ID: P02-4-32	Date: 5/9/02	Prob: 80%	Impact: high

Description:
Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.

Refinement/context:
Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards.
Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components.
Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.

Mitigation/monitoring:
1. Contact third party to determine conformance with design standards.
2. Press for interface standards completion; consider component structure when deciding on interface protocol.
3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.

Management/contingency plan/trigger:
RE computed to be $20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly.
Trigger: Mitigation steps unproductive as of 7/1/02

Current status:
5/12/02: Mitigation steps initiated.

Originator: D. Gagne	Assigned: B. Laster

Exercise

- Refer class notes

- END OF TOPIC